

LD11 Business Interpretation Summary

Detailed business interpretation and recommendation from the extended tuning notebook.

Field	Value
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Submission note
This is an individual PDF document. The documentation package intentionally does not include a combined/merged PDF.

1. Business Meaning of Hyperparameter Tuning

In business analytics, the best model is not always the most complex model. The best model is the one that provides reliable performance, clear interpretation, stable execution, and a strong reason for deployment. Hyperparameter tuning helps compare models more fairly, but the final decision must consider accuracy, ranking power, positive-class detection, runtime, and maintainability.

2. Model Result Interpretation

Model	Result	Business interpretation
Logistic Regression GridSearch	Accuracy 0.9860, F1 0.9889, ROC-AUC 0.9971	Best model for this extended classification workflow. It is strong, simple, explainable, and easy to maintain.
KNN GridSearch	Accuracy 0.9790, F1 0.9836, ROC-AUC 0.9923	Strong baseline, but it depends heavily on distance and scaling. Less practical for very large datasets.
CASH Logistic Regression	Same as Logistic Regression result	Confirms that model family selection prefers Logistic Regression on this dataset.
Hyperopt RF Search	ROC-AUC 0.9933	Advanced search worked, but did not beat the simpler model. Useful for learning and complex future datasets.
Optuna RF Search	ROC-AUC 0.9931	Similar to Hyperopt; strong but not best here.
Successive Halving RF	ROC-AUC 0.9916	Efficient tuning method; useful when compute budget is limited.
Ridge Regression	RMSE 53.2042, R2 0.4881	Shows regression tuning process; performance is moderate because the dataset is difficult for simple linear models.

3. Final Business Recommendation

The final recommended model from the extended classification notebook is Logistic Regression tuned with GridSearchCV. It gives the best metric profile and strongest interpretability. This is a good business outcome because it avoids unnecessary complexity while still producing excellent classification performance.

Hyperopt, Optuna, and Successive Halving should be understood as advanced tools for larger, slower, or more complex projects. They are valuable in production contexts where the search space is bigger and manual tuning becomes inefficient. In this small stable dataset, their practical value is mostly educational and comparative.

Decision statement

Use the simplest model that wins on the important metrics. Here, Logistic Regression wins and is easier to explain than the advanced Random For